

## Research Article

# In Vitro Antiviral Effect of Compound Chinese Herbal Medicine and Probiotic Fermentation Effect on *Siniperca chuatsi*

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*Siniperca chuatsi* is an economically important carnivorous fish. The efficient commercial culture of this fish is hindered by suboptimal nutrient utilization and the emergence of diseases, specifically viral diseases. Compound Chinese herbal medicine (CCHM) and probiotics can be used to enhance disease resistance and improve digestion. We have previously isolated *Bacillus velezensis*, a probiotic with a desirable effect on *S. chuatsi*. In this study, we aimed to evaluate the antiviral effect of a CCHM compound at the cellular level by conducting animal tests to investigate the synergistic effect of CCHM and *B. velezensis* GY65 in a simulated *S. chuatsi* culture. The animal experiments were divided into four groups: control group (G0); CCHM group (G1); *B. velezensis* GY65 group (G2); and Chinese herbal medicine compound probiotic group (CHCP) (G3). CCHM showed significant blocking and killing effects on the largemouth bass virus (LMBV) ( $p < 0.05$ ), while the inhibiting effect was not significant. In animal tests, CCHM significantly improved *S. chuatsi* growth, while CHCP reduced *S. chuatsi* mortality. Regarding biochemical and nonspecific immune parameters, CHCP reduced serum lipid and hepatic malondialdehyde levels and increased serum lysozyme and hepatic catalase activities ( $p < 0.05$ ). CCHM and *B. velezensis* GY65 increased the serum glutathione transaminase levels ( $p < 0.05$ ). Digestive enzymes in the intestine of *S. chuatsi* improved in all groups, and the best result was observed in the CHCP group. Additionally, the intestinal development improved in all groups, but the best intestinal development was observed in the CCHM group. However, in the kidney, only TNF- $\alpha$  was significantly upregulated in the CHCP group ( $p < 0.05$ ). In conclusion, CCHM exerted significant blocking and killing effects on LMBV. CHCP effectively protected the growth and immunity of *S. chuatsi*. Our findings provide preliminary data and lay the foundation for the practical application of herbal medicines and probiotics in aquaculture.

**Keywords:** antiviral effect; compounded Chinese herbs; fermented herbal medicine; mandarin fish growth; probiotics

## 1. Introduction

Mandarin fish (*Siniperca chuatsi*) of the genus *Siniperca*, Perciformes family (Percichthyidae), is of particular economic importance owing to its meat, absence of intermuscular spines, and high nutritional value. However, *S. chuatsi* is a carnivorous fish, and its main food comes from live bait. The feeding nature of *S. chuatsi* has created many problems

in the artificial culture process, such as low bait coefficient, severe water environment pollution, and frequent diseases. This has become a significant obstacle for the large-scale promotion of the mandarin fish artificial culture industry [1]. In addition, the domestication of artificial compound feed to replace live-bait culture is a future direction for the *S. chuatsi* industry. Studies have shown that the use of artificial compound feeds for *S. chuatsi* can reduce bait coefficients, improve

breeding efficiency, and increase profitability. However, this also generates several problems, including reduced disease resistance and difficulties in the digestion and absorption of plant-based feed materials [2]. Consequently, issues on nutritional utilization and emergence of diseases in *S. chuatsi* are eminent in both live bait and feed cultures. In particular, viral diseases highly threaten the health of *S. chuatsi* owing to their increased incidence, rapid prevalence, and lack of effective control measures [3]. Solving these problems has become key to the future development of the mandarin fish industry.

Herbs and probiotics are widely used as feed additives in animal breeding because of their excellent mode of action, greenness, safety, and low side effects [4]. Herbal medicine has been used in China for thousands of years and has its own theoretical basis in Chinese medicine. The application of herbal medicine has been rapidly developed in recent times in combination with modern biotechnology. Chinese herbal medicines are mainly derived from natural plants, which have their own basic nutrients, such as amino acids, carbohydrates, and vitamins, in addition to active ingredients, such as plant polysaccharides, alkaloids, volatile oils, organic acids, and flavonoids [5]. Studies have shown that these active ingredients promote growth and metabolism, improve digestion, and enhance immunity. Herbal medicines also have the advantages of commercialization, such as being natural, non-toxic, inexpensive, and easy to obtain [6]. Herbs have long been widely used as growth promoters or immune enhancers in the aquaculture industry. Probiotics are a class of microorganisms that bring benefits when ingested and promote the health of living organisms, mainly by improving the balance of intestinal microorganisms. Currently, frequently used probiotics in aquaculture include *Bacillus*, *Lactobacillus*, and *Clostridium butyricum* [7], and studies have highlighted their role in improving growth performance, enhancing disease resistance and immunity, improving intestinal microecological balance, and regulating water quality [8]. Herbs and probiotics are similar in terms of their benefits, modes of action, and functions. Most probiotics can produce extracellular enzymes, which allows them to break the walls of herbal medicines, releasing their active ingredients and enhancing their efficacy [9]. Therefore, it is worth exploring whether these compounds can produce a synergistic effect. Many scholars have conducted studies focusing on the synergistic mode, application, and effect between compounds.

Active ingredients, such as polysaccharides, flavonoids, and alkaloids contained in herbs, have antiviral activity and are of interest in the prevention and treatment of viral diseases owing to their low price, low toxicity, and lack of residues [10–12]. Meanwhile, a strain of *Bacillus velezensis* with good extracellular enzyme production ability and biosafety was isolated in our laboratory in the early stage, which promotes the growth of *S. chuatsi*, improves digestion and immunity, and enhances the disease resistance of fish. However, the synergistic application of the compounds in *S. chuatsi* culture has not been reported so far. Therefore, we conducted a study to address this gap.

In this study, a compound Chinese herbal medicine (CCHM) was prepared from *Schisandra chinensis*, *Eucommia*

*ulmoides*, *Taraxacum mongolicum*, *Dichondra micrantha* Urban, *Houttuynia cordata*, *Abrus cantoniensis*, *Smilax glabra* roxb, and *Lotus leaf* based on the principle of CCHM. The compound herbs are rich in polysaccharides, flavonoids, phenols, and other active ingredients, which have hepatoprotective and anti-inflammatory effects and promote immunity [6]. The preliminary experiments showed that this compound herbal medicine exerted an anti-largemouth bass virus (LMBV) effect in vitro and could promote the proliferation of *B. velezensis* GY65. Therefore, we attempted to combine the herbs with probiotics to further develop a feed additive with potential antiviral effects and positive effects on the growth, digestion, and immunity of *S. chuatsi*.

## 2. Materials and Methods

### 2.1. Anti-LMBV Effect of CCHM

**2.1.1. Preparation of LMBV Strain, Endothelial Progenitor Cells (EPCs), and CCHM.** The LMBV strain and EPCs were maintained in our laboratory. The M199 medium, fetal bovine serum, double antibodies, and trypsin were purchased from Gibco, and CCK8 reagent was purchased from Tong Ren Chemical, Japan.

The eight herbs used in this experiment, including *S. chinensis*, *E. ulmoides*, *T. mongolicum*, *D. micrantha* Urban, *H. cordata*, *Abrus cantoniensis*, *S. glabra* roxb, and *Lotus leaf*, were purchased from local pharmacies in Guangzhou. After drying and pulverizing the traditional Chinese medicines, they were passed through an 80-mesh sieve to obtain fine powders, which were mixed in a respective mass ratio of 1 : 1 : 1 : 1 : 2 : 2 : 2 : 2 to obtain a CCHM preparation. The mixed compound herbs were then wrapped in absorbent cotton gauze, added to five times the volume of water, soaked for 0.5 h, boiled, decocted for 0.5 h, filtered through gauze, and the residue was decocted again. The filtrate was mixed, evaporated, and concentrated until the effective drug concentration was 1 g/mL. After boiling and sterilization, it was stored at 4°C until further use.

**2.1.2. Determination of LMBV Titers.** The virus to be tested was diluted 10-fold and added to 96-well plates containing EPC cells at a dose of 100 µL/well, with eight replicates of each dilution. A normal cell control was used as a negative control. After 1.5 h of adsorption at 28°C, the virus solution was discarded, a cell maintenance solution containing 5% serum was added and placed in a 5% CO<sub>2</sub> incubator for 7 days of continuous observation, and half of the infectious dose in tissue culture was calculated according to Karber's method [13, 14].

**2.1.3. Determination of Safe CCHM Concentration in EPCs.** The maximum safe concentration of CCHM in EPCs was determined using the cloud point extraction (CPE) method. The drug was filtered through a 0.22 mm sterile filter membrane and inoculated into 96-well plates filled with monolayer EPCs after multiplicative dilution, and four replicates of each dilution were made. Another group of normal cells was incubated at 28°C for 96 h, and the maximum concentration

that did not cause cytopathy was recorded as the safe concentration of the drug.

**2.1.4. Determination of the Blocking Effect of CCHM Against LMBV.** Five CCHM dilutions of 12.5, 6.25, 3.125, 1.5625, and 0.78125 mg/mL were prepared by multiplicative dilution from a safe concentration (12.5 mg/mL) and added to a 96-well plate containing a single EPC layer. Each drug concentration (100  $\mu$ L) was added, and four replicate wells were prepared for each dilution. After incubation at 28°C for 4 h, each well was aspirated, and 100  $\mu$ L of 100 TCID<sub>50</sub> viral solution was added and adsorbed at 28°C for 1.5 h, followed by aspiration of viral solution and addition of cell maintenance solution; new cell and viral controls were set up. A cell incubator at 28°C was used, and CPE was observed daily; when the viral control CPE reached “+++”++++” (when the CPE reaches “+++”++++” the cytopathic effect has reached “75%”100%”), CPE was recorded, and the survival rate (SR) of cells in the 96-well plates was determined using the CCK8 method.

**2.1.5. Determination of the Inhibitory Effect of CCHM Against LMBV.** After the EPCs in the 96-well cell culture plate grew into a monolayer, the virus solution was inoculated with 100 TCID<sub>50</sub> at a spiked amount of 100  $\mu$ L per well and adsorbed at 28°C for 1.5 h. The virus solution was discarded, and CCHM was added at a safe concentration range of 100  $\mu$ L per well. Four well replicates for each dilution were prepared, and new cell and virus controls were set up. A cell culture incubator at 28°C was used to incubate the cells. The CPE was observed daily, and when the virus control CPE reached “+++”++++” the CPE was recorded, and the SR of cells in the 96-well plates was determined using the CCK8 method.

**2.1.6. Determination of the Killing Effect of CCHM Against LMBV.** Five CCHM dilutions in the safe concentration range were mixed with an equal volume of 100 TCID<sub>50</sub> virus solution and incubated at 28°C for 2 h. The CCHM solution was then added into 96-well cell culture plates with monolayer EPCs at 100  $\mu$ L per well; four well replicates were prepared for each dilution, and new cell and virus controls were set up. A cell culture incubator at 28°C was used to observe the CPE daily, and when the virus control CPE reached “+++”++++” the CPE was recorded, and the SR of cells in the 96-well plates was determined using the CCK8 method.

**2.2. Preparation of Feed Additives.** According to the aforementioned method, 1 kg of compound herbs was extracted by decoction, concentrated, and adjusted to 1000 mL with sterile water. According to the results of in vitro antiviral test and growth pretest, 2 g of compound Chinese herbs per kg of basal diet was selected as the experimental concentration, which is 2 mL of CHMM per kg of basal diet.

*B. velezensis* GY65 used in this experiment was kept by the Pearl River Institute of Aquatic Sciences, Chinese Academy of Aquatic Sciences. Before the experiment, *B. velezensis* GY65 was inoculated into LB solid medium for activation, and the activated *B. velezensis* GY65 was inoculated into LB liquid medium and incubated at 37°C using a shaker at 200 rpm until logarithmic growth phase (OD<sub>600</sub> = 0.6). Subsequently, 20% of the total volume was inoculated into 1 g/mL of the herbal

concentrate, and the number of GY65 in the fermented herbal solution was determined to be  $5 \times 10^8$  cfu/mL after 24 h of incubation at 37°C on a shaker at 200 rpm by counting the concentration of live bacteria in the fermented herbal solution at different incubation times by plate counting method, which means that the number of live bacteria was  $1 \times 10^9$  cfu/kg of feed added to 2 mL of the fermented herbal solution.

Activated *B. velezensis* GY65 was inoculated into LB liquid medium for 24 h at 37°C for enrichment, centrifuged at 5500 rpm for 10 min at 4°C, and then resuspended and collected in 0.65% saline. Subsequently, to bring the viable counts to feeding levels, we calculated the viable counts using the plate count method and adjusted the concentration to  $5 \times 10^8$  cfu/mL with normal saline, adding 2 mL/kg of feed for a total viable count of  $1 \times 10^9$  cfu.

CCHM, GY65 was diluted with sterile water and sprayed evenly on the surface of the basal diet and left at room temperature for 1 h to allow full absorption of the feed. The prepared feeds were stored at –20°C and restored to room temperature before feeding. The basal diet was specially designed for *S. chuatsi* without the addition of non-nutritional ingredients such as probiotics, herbs, and antibacterial substances, and its nutritional composition is shown in Table 1. The composition of feed additives for each group is also shown in Table 1.

**2.3. Preparation and Feeding Management of Fish.** Animal experiments were conducted in ponds at the Pearl River Fisheries Research Institute of the Chinese Academy of Fishery Sciences. After temporary storage for 7 days, 450 healthy mandarin fish ( $29.4 \text{ g} \pm 0.2 \text{ g}$ ) that had been domesticated and fed an artificial diet were randomly assigned to 12 cages ( $1.3 \times 1.5 \times 1.3 \text{ m}$ ). The experiment consisted of one control group and three experimental groups, each with three replicates. Breeding lasted 8 weeks, and the fish were fed twice daily (6:30 and 18:30) at 1%–3% of body weight. During the breeding period, the water temperature varied from 14.1 to 24.3°C, and the ammoniacal nitrogen and ammonium nitrite contents were consistent with the water quality of the pond.

**2.4. Sample Collection.** The fish were weighed, and their body lengths were measured before the test. After the 56th day of the trial, samples were collected after the feeding was stopped for 1 day. During sampling, 12 fish were randomly selected from each group, with four replicates each. Tail vein blood was collected after weighing and measuring the body length. Blood was stored overnight at 4°C. Whole-blood samples were centrifuged at 4000 rpm for 10 min at 4°C, and serum of five fish was collected and stored in a –80°C refrigerator until detection. After blood collection, the visceral mass and the liver were removed and weighed. The spleen and kidney tissues of four fish were collected and placed in a 2 mL centrifuge tube containing 1 mL of Trizol to determine the relative mRNA expression of immune-related genes. The liver tissue of five fish was placed in a 2 mL cryopreservation tube to determine nonspecific immune enzyme activity. The mid-hindgut of five fish was washed with sterile water and placed in a 2 mL cryopreservation tube to determine digestive enzyme activity. All samples were stored at –80°C until detection. The midhindgut of three fish from each group

TABLE 1: Nutrient composition of basal diets.

| Items (%)                                         | Control | G1     | G2     | G3     |
|---------------------------------------------------|---------|--------|--------|--------|
| <b>Ingredients</b>                                |         |        |        |        |
| Fish meal                                         | 42.00   | 42.00  | 42.00  | 42.00  |
| High gluten flour                                 | 12.80   | 12.80  | 12.80  | 12.80  |
| Dried porcine soluble                             | 8.00    | 8.00   | 8.00   | 8.00   |
| Soybean meal                                      | 9.00    | 9.00   | 9.00   | 9.00   |
| Peanut meal                                       | 8.00    | 8.00   | 8.00   | 8.00   |
| Meat and bone meal                                | 5.00    | 5.00   | 5.00   | 5.00   |
| Blood meal                                        | 3.00    | 3.00   | 3.00   | 3.00   |
| Beer yeast                                        | 2.00    | 2.00   | 2.00   | 2.00   |
| Fish oil                                          | 2.00    | 2.00   | 2.00   | 2.00   |
| Squid paste                                       | 2.00    | 2.00   | 2.00   | 2.00   |
| Soybean oil                                       | 2.00    | 2.00   | 2.00   | 2.00   |
| Soybean phospholipids                             | 2.00    | 2.00   | 2.00   | 2.00   |
| Ca (H <sub>2</sub> PO <sub>4</sub> ) <sub>2</sub> | 1.00    | 1.00   | 1.00   | 1.00   |
| Choline chloride                                  | 0.30    | 0.30   | 0.30   | 0.30   |
| Glycine                                           | 0.10    | 0.10   | 0.10   | 0.10   |
| Taurine                                           | 0.20    | 0.20   | 0.20   | 0.20   |
| Yeast nucleotide                                  | 0.30    | 0.30   | 0.30   | 0.30   |
| Vitamin premix <sup>a</sup>                       | 0.10    | 0.10   | 0.10   | 0.10   |
| Mineral premix <sup>b</sup>                       | 0.20    | 0.20   | 0.20   | 0.20   |
| Total                                             | 100.00  | 100.00 | 100.00 | 100.00 |
| <b>Feed additives (per kg of basal diet)</b>      |         |        |        |        |
| 0.65% normal saline (mL)                          | 2.00    | 0.00   | 0.00   | 0.00   |
| CHMM (mL)                                         | 0.00    | 2.00   | 0.00   | 0.00   |
| GY65 bacteria liquid (cfu)                        | 0.00    | 0.00   | 2.00   | 0.00   |
| CHCP (mL)                                         | 0.00    | 0.00   | 0.00   | 2.00   |
| <b>Nutrient levels (%)<sup>c</sup></b>            |         |        |        |        |
| Crude protein                                     | 37.89   | 37.96  | 37.87  | 37.81  |
| Ether extract                                     | 7.62    | 7.64   | 7.61   | 7.59   |
| Moisture                                          | 27.54   | 27.49  | 27.55  | 27.44  |
| Ash                                               | 7.42    | 7.47   | 7.40   | 7.45   |

<sup>a</sup>One kilogram of vitamin premix contain the follow: VA, 60,606,060.6 IU; VD, 400,000,000 IU; VE, 1497 IU; VK, 2 g; VB<sub>1</sub>, 5 g; VB<sub>2</sub>, 5 g; VB<sub>6</sub>, 5 g; nicotinic acid, 20 g; pantothenate, 20 g; biotin, 1 g; folic acid, 10 g; VB<sub>12</sub>, 5 g; and defatted rice bran, 900 g.

<sup>b</sup>One kilogram of mineral premix contain the follow: CuCO<sub>3</sub>, 4 g; FeC<sub>6</sub>H<sub>5</sub>O<sub>7</sub>, 15 g; MgO, 25 g; MnSO<sub>4</sub>, 6 g; KCl, 240 g; ZnSO<sub>4</sub>, 60 g; NaCl, 50 g; and zeolite powder, 600 g.

<sup>c</sup>Nutrient levels were measured values.

was stored in a centrifuge tube containing 1 mL of neutral tissue fixative for the production of intestinal tissue slices.

**2.5. Growth Performance.** Growth performance parameters, such as weight gain rate (WGR), specific growth rate (SGR), SR, viscerosomatic index (VSI), hepatosomatic index (HSI), and condition factor (CF) were calculated as follows:

$$\text{Weight gain rate (\%)} = 100 \times (W_f - W_i) / W_i,$$

$$\text{Specific growth rate (\% day}^{-1}\text{)} = 100 \times (\ln W_f - \ln W_i) / t,$$

$$\text{Survival rate (\%)} = 100 \times N_t / N_0,$$

$$\text{Viscerosomatic index (\%)} = 100 \times W_v / W_f,$$

$$\text{Hepatosomatic index (\%)} = 100 \times W_h / W_f,$$

$$\text{Condition factor (g/cm}^3\text{)} = W_f / (W_f)^3,$$

where  $W_f$  and  $W_i$  are the final and initial fish body weights, respectively;  $t$  is the experimental duration in days;  $N_0$  and  $N_t$  are the number of fish at the start and end of the trials, respectively;  $W_v$  and  $W_h$  are the visceral and liver weights, respectively; and  $L_f$  is the initial fish body length.

**2.6. Serum and Liver Biochemical Indicators and Nonspecific Immune Analysis.** Total cholesterol (T-CHO), triglycerides (TGs), lysozyme (LZM), superoxide dismutase (SOD), catalase (CAT), malondialdehyde (MDA), glutamic oxaloacetic transaminase (GOT), and glutamic pyruvic transaminase (GPT) were determined using kits from the Nanjing Jiancheng Biological Engineering Institute.

**2.7. Intestinal Digestive Enzyme Activity.** Intestine or liver (0.1 g) were collected from each mandarin fish, and 1 mL of phosphate-buffered saline and several magnetic bead homogenates were added to prepare a tissue suspension. The supernatant was transferred to a clean 10 mL tube at 4°C and centrifuged at 6100 rpm for 15 min for the quantitative determination of amylase and protease activities.

The intestinal or liver supernatant (1 mL) was mixed with 5 mL of 0.7% (W/V) casein solution at 37°C for 15 min. Subsequently, 3 mL of 110 mmol/L trichloroacetic acid was added, and the supernatant was centrifuged at 4°C at 12,000 rpm for 15 min. The supernatant was transferred to a new 10 mL tube, and 1 mL of 0.5 mmol/L florin phenol and 5 mL of 500 mmol/L florin phenol were added. The supernatant was incubated at 37°C for 30 min, with tyrosine as the standard. Absorbance was measured at 440 nm using a UV-visible spectrophotometer, and a calibration curve was constructed. The amount of tyrosine released was 1 µmol/L per min. The amount of enzyme released was considered to be one unit of protease activity [15].

The supernatant from the intestine or liver (0.5 mL) was mixed with 0.5 mL of 1% soluble amylose solution and incubated for 30 min at 37°C. Subsequently, 1 mL of DNS reagent was added, boiled for 5 min, and cooled to ~25°C. The absorption was recorded at 540 nm using a UV-visible spectrophotometer. The standard curve was established using glucose as the standard, and one unit of amylase activity was considered 1 µg of reducing sugar per min. Lipase activity was measured using a kit from the Nanjing Jiancheng Institute of Biological Engineering.

**2.8. Intestinal Histological Assessment.** Midhindgut tissue was fixed in Bouin's solution for 24 h, dehydrated, embedded in paraffin, and cut into 4-µm transverse incisions along the axis of the intestinal lumen. Hematoxylin and eosin staining was performed, and the samples were histologically examined using a light microscope (Nikon, Tokyo, Japan) and a digital camera (Nikon, Tokyo, Japan). The height of the intestinal villi and the thickness of the basement

TABLE 2: Sequences of primer pairs used for real-time quantitative PCR in this study.

| Primer name      | Primer sequences (5'~3') | Annealing temperature (°C) | Accession number |
|------------------|--------------------------|----------------------------|------------------|
| $\beta$ -Actin-F | GGAGAAGCTGTGGTACGTCGC    | 60                         | AY885683.1       |
| $\beta$ -Actin-R | TCTGGGCAACGGAACCTCTCA    |                            |                  |
| TNF- $\alpha$ -F | CGGGGCTCCAGTTTTCTCYG     | 60                         | AY839276.1       |
| TNF- $\alpha$ -R | ATGTCCACAGCAGAGGCGTTG    |                            |                  |
| IL-8-F           | GGGATGAGCCTGAGAAGCCTG    | 60                         | KP409192.1       |
| IL-8-R           | GGCAGGAATCAGCTCCACCTT    |                            |                  |

Abbreviation: PCR, polymerase chain reaction.

membrane of each segment were measured using the Image-Pro software.

**2.9. Gene Relative Expression Analysis.** Total RNA was extracted from the posterior kidney and spleen tissue stored at  $-80^{\circ}\text{C}$  using TRIzol reagent (Thermo, USA) according to the manufacturer's protocol. The RNA quality was evaluated using agarose gel electrophoresis, and RNA concentration was determined using spectrophotometry (Thermo, USA). Approximately 1  $\mu\text{g}$  of total RNA from each of tissue samples was subjected to a reverse transcription (RT) kit (Takara, Japan) according to the manufacturer's instructions to obtain cDNA. The cDNA samples were stored at  $-20^{\circ}\text{C}$  until the quantitative real-time polymerase chain reaction (PCR) reaction was used. The expression levels of TNF- $\alpha$ , IL-8, and  $\beta$ -actin were determined using a quantitative PCR. The expression levels of  $\beta$ -actin were used as an internal control. The specific primers used for detecting each gene are listed in Table 2. RT-PCR was performed on a real-time fluorescence quantitative PCR instrument (ABI 7500, USA) with a TransGen Biotech SYBR RT-qPCR kit (Beijing, China). Fluorescence quantitative PCR reaction system:  $2 \times$  TransStart Top Green qPCR SuperMix, 10  $\mu\text{L}$ ; upstream primer 0.4  $\mu\text{L}$ ; downstream primer 0.4  $\mu\text{L}$ ; cDNA template 2  $\mu\text{L}$ ; RNase Free dH<sub>2</sub>O, 6.8  $\mu\text{L}$ ; ROX 1.0  $\mu\text{L}$ ; total 20  $\mu\text{L}$ . PCR conditions were as follows:  $94^{\circ}\text{C}$  for 30 s; 40 circles of  $94^{\circ}\text{C}$ ,  $60^{\circ}\text{C}$ , and  $72^{\circ}\text{C}$  for 5, 15, and 10 s, respectively. Relative expression levels of each group were normalized to  $\beta$ -actin and are expressed as the mean  $\pm$  standard error.

**2.10. Statistical Analysis.** One-way statistical analysis of the data was performed using SPSS (IBM, Armonk, New York). Subsequently, differences among means were identified by Tukey's post hoc test and a considered significant at  $p < 0.05$ . The  $2^{-\Delta\Delta\text{CT}}$  method was used to calculate the relative expression of immune-related genes, which was defined as 1 for the control group at different time points.

### 3. Results

**3.1. LMBV Titers and Safe Drug Concentrations for CCHM.** After 7 days of observation, the LMBV titer was determined to be  $10^{-4.48}$  TCID<sub>50</sub>/mL. In addition, in order to avoid the interference of CCHM toxicity in the subsequent experiments, the safe concentration of CCHM in EPC was determined to be 12.50 mg/mL by using the CPE method.

**3.2. Blocking, Inhibiting, and Killing Effects of CCHM Against LMBV.** In assessing the blocking effect, the cell SRs at CCHM concentrations of 6.25 and 12.5 mg/mL were significantly higher than those of the virus control group ( $p < 0.05$ ), and there was a gradual increase in the cell SR with increasing concentration. When evaluating the killing effect, the cell SRs at CCHM concentrations  $> 0.78125$  mg/mL were significantly higher ( $p < 0.05$ ) than those of the control group, and the rate increased with increasing concentrations. However, the inhibitory effect of CCHM on the virus was not significant, and the difference between the groups was also not significant ( $p > 0.05$ ) (Figure 1 and Table 3).

**3.3. Growth Performance.** Growth performance was assessed after 56 days of adding equal amounts of equivalent herbal mixtures, probiotics, and probiotic fermented herbs to mandarin fish feed. Adding Chinese herbs to the diet significantly increased the WGR of *S. chuatsi*, compared to that in the control group ( $p < 0.05$ ) (Table 4). The experimental group showed some improvement in SGR, SR, VSI, HSI, and CF compared to the control group; however, the difference was not significant ( $p > 0.05$ ). Unexpectedly, the SR of mandarin fish was relatively low in G1, while the SR in G3 showed good improvement, compared to the other groups.

**3.4. Biochemical and Nonspecific Immune Parameters of the Serum.** Observing and comparing the results of serum indices in the different *S. chuatsi* groups showed a significant decrease in the serum TG content in each experimental group ( $p < 0.05$ ), relative to the control group. Among them, the TG content in the serum of G3 was the lowest (Figure 2B). LZM activity in the serum of all experimental groups was significantly higher than that of the control group ( $p < 0.05$ ), with the highest activity observed in G3 (Figure 2C). Serum GOT activity was significantly higher in the group fed with a diet supplemented with herbal medicine than in the control and G3 groups ( $p < 0.05$ ) (Figure 2E). There were no significant differences in serum T-CHO, as well as CAT and GPT activities between the groups ( $p > 0.05$ ) (Figure 2).

**3.5. Biochemical and Nonspecific Immune Parameters of the Liver.** The biochemical and nonspecific immune indices of the liver showed that CAT activity was significantly higher in the experimental group than in the control group ( $p < 0.05$ ). CAT activity in the G1 group was significantly higher than that in the other groups ( $p < 0.05$ ) (Figure 3B). Conversely, the MDA content of the experimental group was lower than

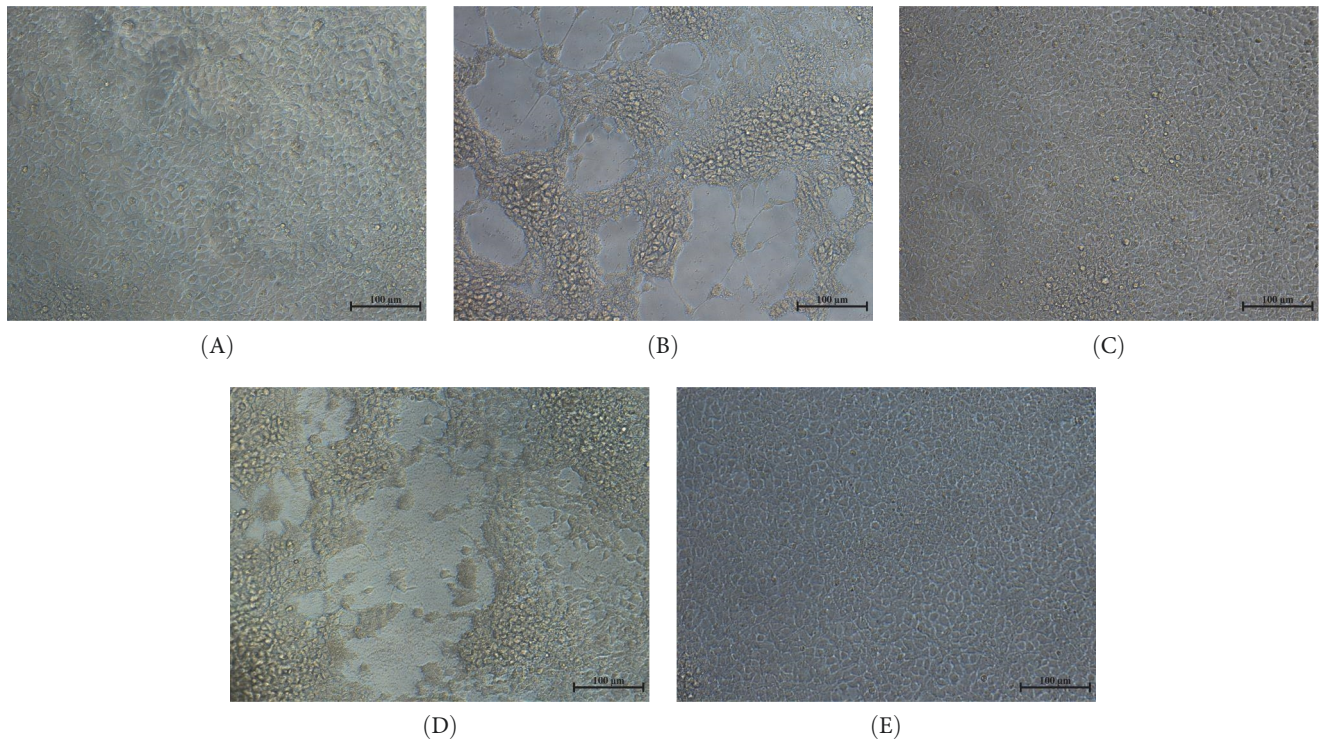


FIGURE 1: EPC cell state under the blocking, inhibiting, and killing effects of CHMM at a concentration of 12.5 mg/mL against LMBV: (A) untreated control, (B) virus control, (C) blocking effect, (D) inhibiting effect, and (E) killing effect. EPC, endothelial progenitor cell; LMBV, largemouth bass virus.

TABLE 3: Determination of the blocking effect, inhibiting effect, and killing effect of CCHM against LMBV.

| CCHM concentration (mg/mL) | Control                   | 0.78125                    | 1.5625                     | 3.125                        | 6.25                          | 12.5                       |
|----------------------------|---------------------------|----------------------------|----------------------------|------------------------------|-------------------------------|----------------------------|
| Blocking effect            | 38.40 ± 1.19 <sup>b</sup> | 25.42 ± 2.60 <sup>a</sup>  | 37.93 ± 2.01 <sup>b</sup>  | 49.14 ± 10.80 <sup>b</sup>   | 79.53 ± 6.67 <sup>c</sup>     | 115.65 ± 9.57 <sup>d</sup> |
| Cell viability (%)         |                           |                            |                            |                              |                               |                            |
| Inhibiting effect          | 38.40 ± 1.19              | 33.71 ± 1.56               | 26.68 ± 1.65               | 31.56 ± 0.93                 | 26.63 ± 0.44                  | 35.56 ± 19.23              |
| Killing effect             | 38.40 ± 1.19 <sup>a</sup> | 105.56 ± 9.27 <sup>b</sup> | 110.25 ± 8.02 <sup>b</sup> | 116.43 ± 8.22 <sup>b,c</sup> | 121.82 ± 14.55 <sup>b,c</sup> | 131.83 ± 7.85 <sup>c</sup> |

Note: Data are presented as mean ± SD. Mean values in the same row with same letters are not significantly different ( $p > 0.05$ ), whereas different letters within a column represent a significant difference ( $p < 0.05$ ).

Abbreviations: CCHM, compound Chinese herbal medicine; LMBV, largemouth bass virus.

TABLE 4: Growth performance and survival of *S. chuatsi* after consuming food containing herbal mixtures (G1), probiotics (G2), and probiotic fermented Chinese herbs (G3) for 56 days, respectively.

| Group       | Control                     | G1                          | G2                           | G3                           |
|-------------|-----------------------------|-----------------------------|------------------------------|------------------------------|
| WGR (%)     | 104.73 ± 17.14 <sup>a</sup> | 121.31 ± 11.10 <sup>b</sup> | 109.53 ± 9.18 <sup>a,b</sup> | 107.96 ± 3.13 <sup>a,b</sup> |
| SGR (%/day) | 1.28 ± 0.15                 | 1.42 ± 0.09                 | 1.32 ± 0.08                  | 1.31 ± 0.02                  |
| SR (%)      | 95.56 ± 4.16                | 90.00 ± 7.20                | 96.67 ± 4.71                 | 100.00 ± 0.00                |
| VSI (%)     | 6.79 ± 0.53                 | 6.87 ± 0.69                 | 7.66 ± 0.77                  | 7.60 ± 0.93                  |
| HSI (%)     | 1.53 ± 0.12                 | 1.51 ± 0.12                 | 1.72 ± 0.35                  | 1.56 ± 0.37                  |
| CF (%)      | 2.91 ± 0.17                 | 2.89 ± 0.23                 | 3.01 ± 0.33                  | 2.96 ± 0.14                  |

Note: Data are presented as mean ± SD ( $n = 12$ ). Mean values in the same row with same letters are not significantly different ( $p > 0.05$ ), whereas different letters within a column represent a significant difference ( $p < 0.05$ ).

Abbreviations: CF, condition factor; HSI, hepatosomatic index; SGR, specific growth rate; SR, survival rate; VSI, viscerosomatic index; WGR, weight gain rate.

that of the control group, and the G3 group results were significantly lower ( $p < 0.05$ ) (Figure 3C). In addition, there were no significant differences in GPT and GOT activities in the liver in each group ( $p > 0.05$ ) (Figure 3).

**3.6. Digestive Enzyme Activities.** The three intestinal digestive enzyme activities in the experimental groups increased compared to those in the control group (Figure 4). In particular, the protease and lipase activities in G1 and G3 were significantly

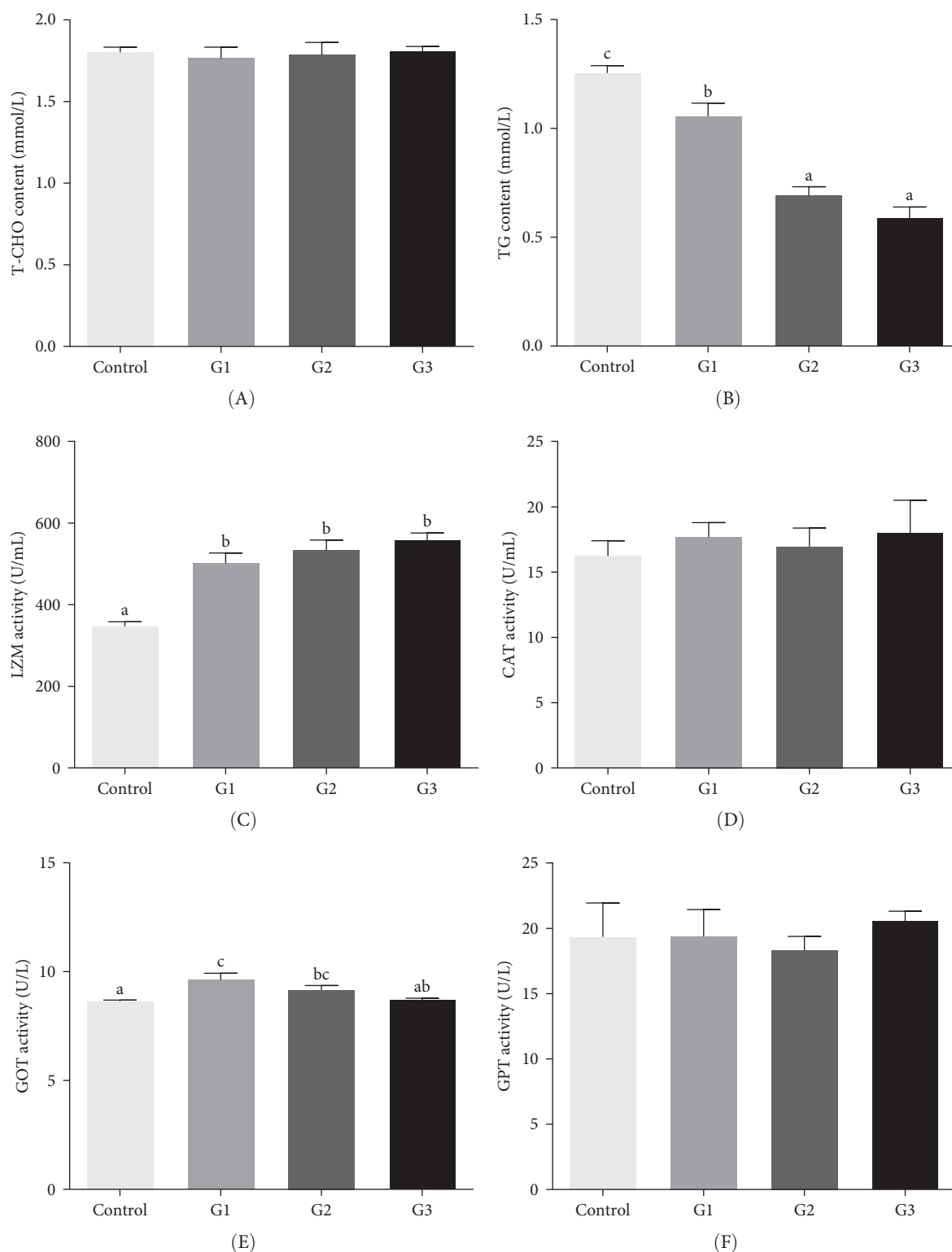


FIGURE 2: Biochemical and nonspecific immunological markers of serum, including total cholesterol (A), total triglycerides (B), lysozyme activity (C), catalase activity (D), glutamic oxalacetic transaminase activity (E), and glutamic pyruvic transaminase activity (F) are indicated. Data are presented as mean  $\pm$  SD ( $n = 5$ ). Bars with different letters represent significant differences ( $p < 0.05$ , Tukey's test).

higher than those of the control group ( $p < 0.05$ ) (Figure 4A, C), and the amylase activities in G2 and G3 were significantly higher than that of the control and G1 groups ( $p < 0.05$ ) (Figure 4B).

**3.7. Intestinal Morphology.** The intestinal development in all experimental groups was better than that in the control group (Figure 5 and Table 5). The length of the villi in all groups was significantly greater than that in the control

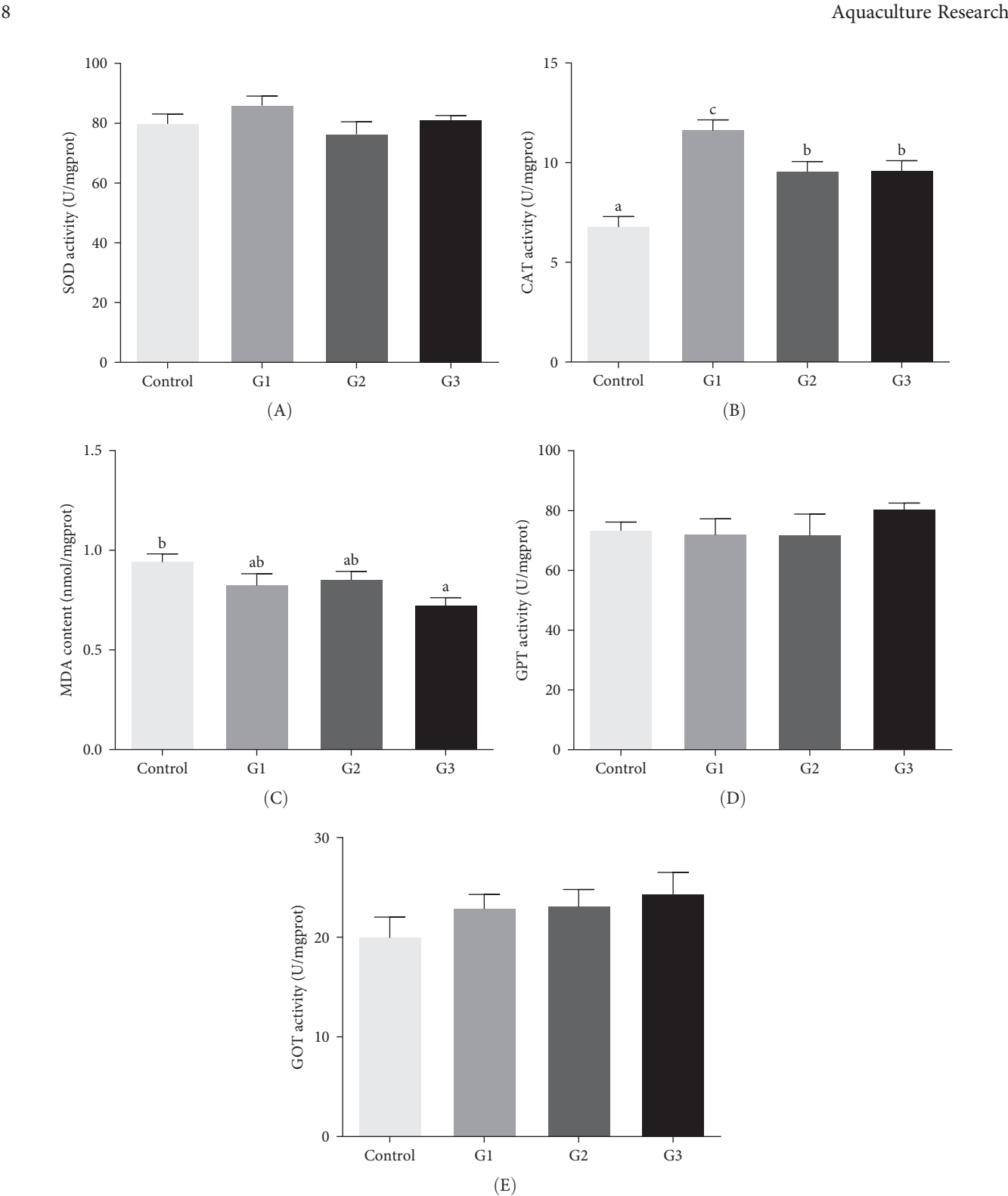


FIGURE 3: Biochemical and nonspecific immunological markers of the liver, including superoxide dismutase activity (A), catalase activity (B), malondialdehyde content (C), glutamic oxalacetic transaminase activity (D), and glutamic pyruvic transaminase activity (E) are indicated. Data are presented as mean  $\pm$  SD ( $n = 5$ ). Bars with different letters represent significant differences ( $p < 0.05$ , Tukey's test).

group ( $p < 0.05$ ), with G1 having the longest villi. The width of the villi in G2 was significantly greater than that in the control group ( $p < 0.05$ ), and no significant difference was observed in the remaining groups ( $p > 0.05$ ). The thickness

of the muscular layer was significantly higher in each experimental group, compared to the control group, and G1 thickness was significantly higher than that in the remaining groups ( $p < 0.05$ ).

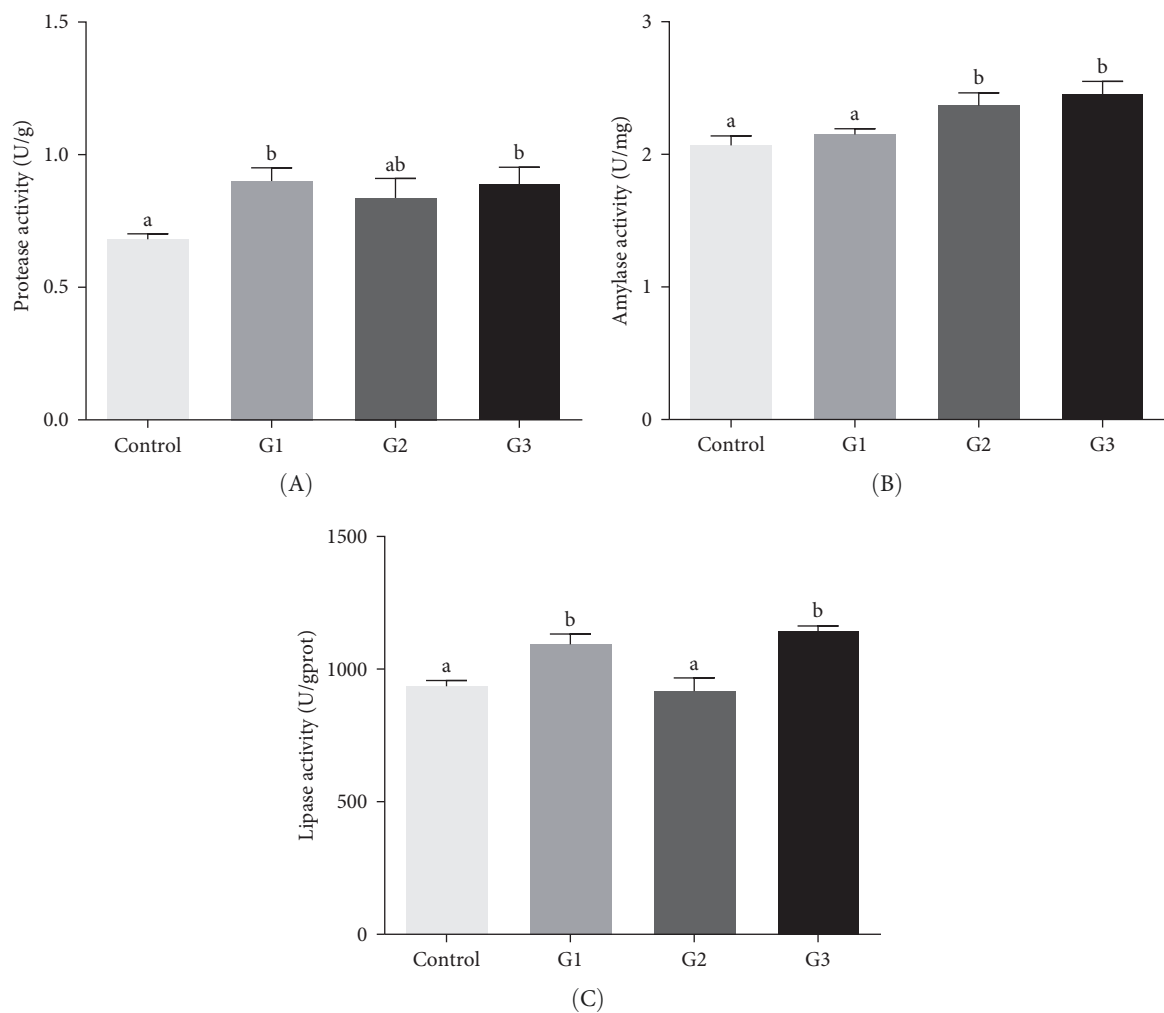


FIGURE 4: (A–C) Digestive enzyme activities in the intestine of *S. chuatsi*. Data are presented as mean  $\pm$  SD ( $n = 5$ ). Bars with different letters represent significant differences ( $p < 0.05$ , Tukey's test).

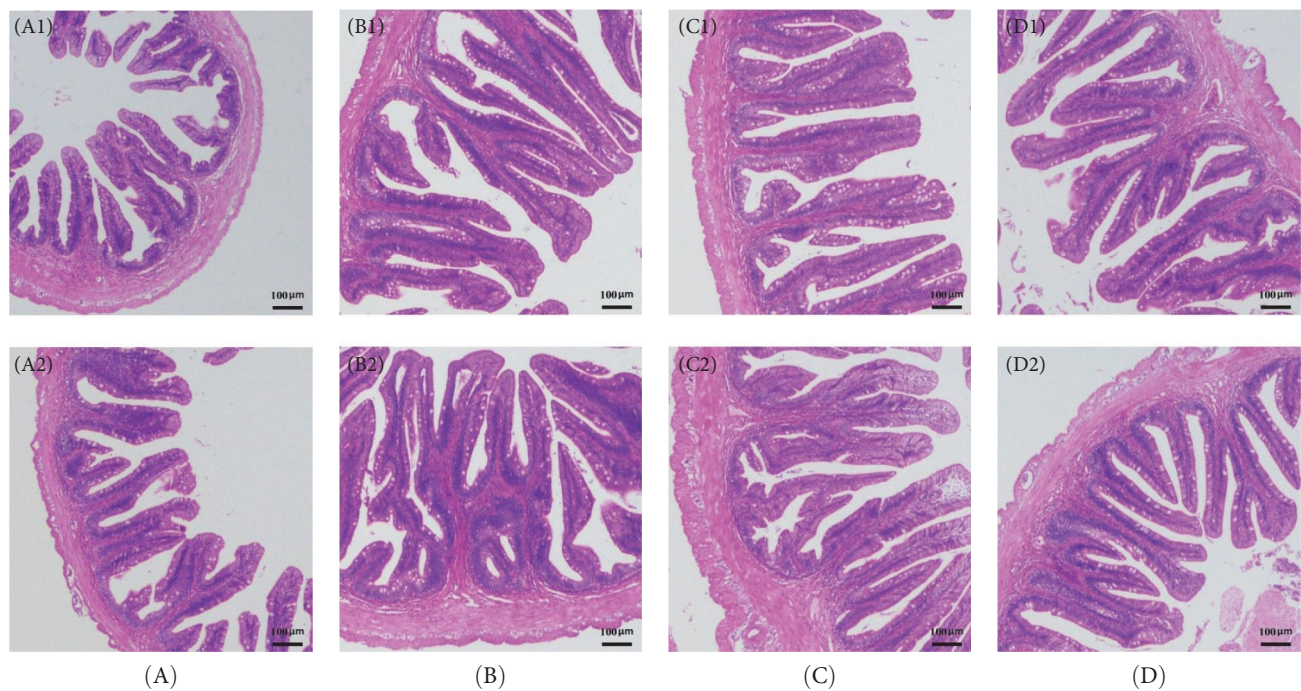


FIGURE 5: Histomorphology of the intestine of *S. chuatsi*: (A) control (A1, A2), (B) G1 (B1, B2), (C) G2 (C1, C2), and (D) G3 (D1, D2).

TABLE 5: Histomorphometric measurements of the mid and posterior intestine of *S. chuatsi* after 56 days of consuming food containing herbal mixtures, probiotics, and probiotic fermented herbs, respectively.

| Group (μm) | Control                     | G1                            | G2                          | G3                            |
|------------|-----------------------------|-------------------------------|-----------------------------|-------------------------------|
| VH         | 457.62 ± 35.08 <sup>a</sup> | 705.69 ± 58.26 <sup>c</sup>   | 682.36 ± 49.46 <sup>c</sup> | 597.83 ± 29.02 <sup>b</sup>   |
| VT         | 97.70 ± 16.75 <sup>a</sup>  | 106.74 ± 17.58 <sup>a,b</sup> | 139.16 ± 25.08 <sup>b</sup> | 120.34 ± 20.37 <sup>a,b</sup> |
| MLT        | 46.48 ± 6.03 <sup>a</sup>   | 89.01 ± 2.25 <sup>c</sup>     | 68.71 ± 10.20 <sup>b</sup>  | 70.48 ± 6.25 <sup>b</sup>     |

Note: Data are presented as mean ± SD (*n* = 3). Mean values in the same row with same letters are not significantly different (*p* > 0.05), whereas different letters within a column represent a significant difference (*p* < 0.05). Abbreviations: MLT, muscularity layer thickness; VH, villus height; VT, villus thickness.

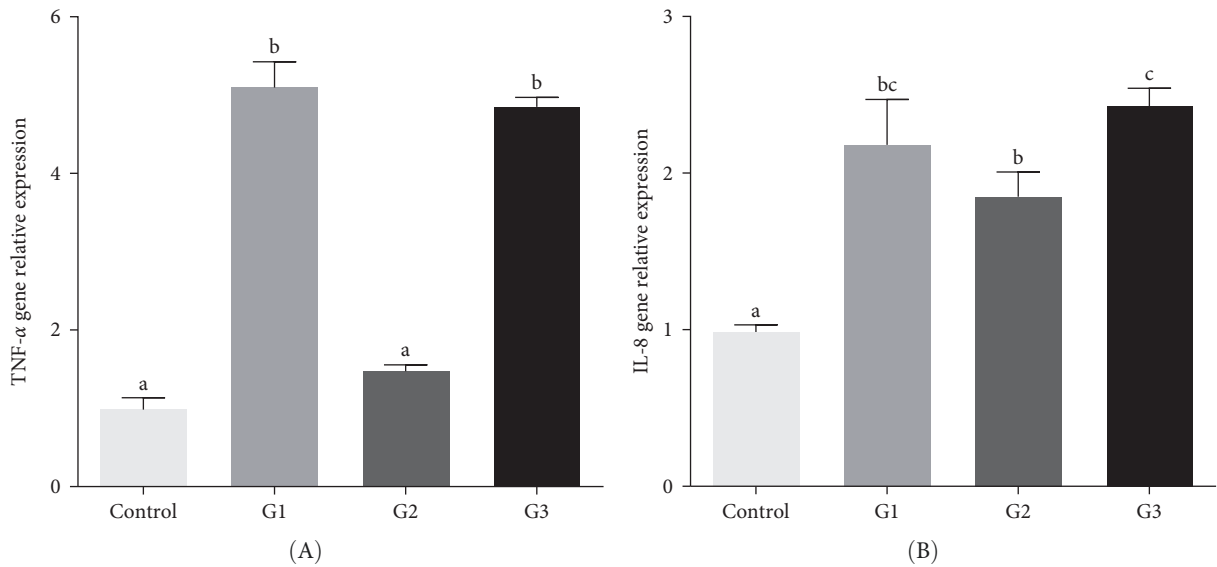


FIGURE 6: Gene relative expression of TNF- $\alpha$  (A) and IL-8 (B) in the spleen of *S. chuatsi*. Data are presented as mean  $\pm$  SD (*n* = 4). Bars with different letters represent significant differences (*p* < 0.05, Tukey's test).

**3.8. Related Immune Gene Expression.** The relative expression of TNF- $\alpha$  and IL-8 in the spleen of the mandarin fish across the experimental groups was greater than that in the control group (Figure 6), and their expression was significantly greater in G1 and G3 than that in the control group (*p* < 0.05) (Figure 6A). The relative TNF- $\alpha$  expression was significantly greater in G3 than in the other groups in the kidney (*p* < 0.05) (Figure 7A). The relative IL-8 expression was not significantly different between the groups (*p* > 0.05) (Figure 7B).

4. Discussion

Currently, the antiviral mechanisms of herbal medicine are classified into direct and indirect antiviral effects, according to the mode of action. Direct antiviral effects can be divided into blocking, inhibiting, and killing effects. Conversely, indirect antiviral effects occur mainly via immune system activity in the body. In this study, we investigated the direct antiviral effect of CCHM through a twofold dilution based on its safe concentration (12.5 mg/mL) for cells and found that CCHM exerted good blocking and killing effects on LMBV, which became stronger with increasing drug concentration. However, CCHM did not have a significant

inhibitory effect on the virus, consistent with the results of previous studies. *H. cordata* has significant direct antiviral effects against a variety of viruses, with flavonoid components playing a major role [7, 11]. The active antiviral components act by interfering with the process of viral infestation of the cells [16, 17], inhibiting protein expression, and interfering with viral proliferation [18]. The active ingredients in *A. cantoniensis*, *L. leaf*, and *T. mongolicum* also exert antiviral effects by directly killing the virus or inhibiting viral replication [12, 19, 20]. In the present study, the antiviral effect of CCHM on LMBV was evaluated in terms of killing, blocking, and inhibitory effects, showing a decrease in superiority in this order. The results of the study could be attributed to the antiviral mechanism wherein the active ingredients of Chinese herbs act in the cellular phase of virus adsorption and proliferation. However, we did not conduct a specific investigation to determine which herbal medicine components exerted antiviral effects in this study, which is one of our future research directions. We will also conduct further research on the underlying mechanism of the antiviral activity of the herbs.

In animal tests, the evaluation of common growth indicators, such as weight gain, specific growth, and SRs, can accurately reflect the direct breeding effect of each group of

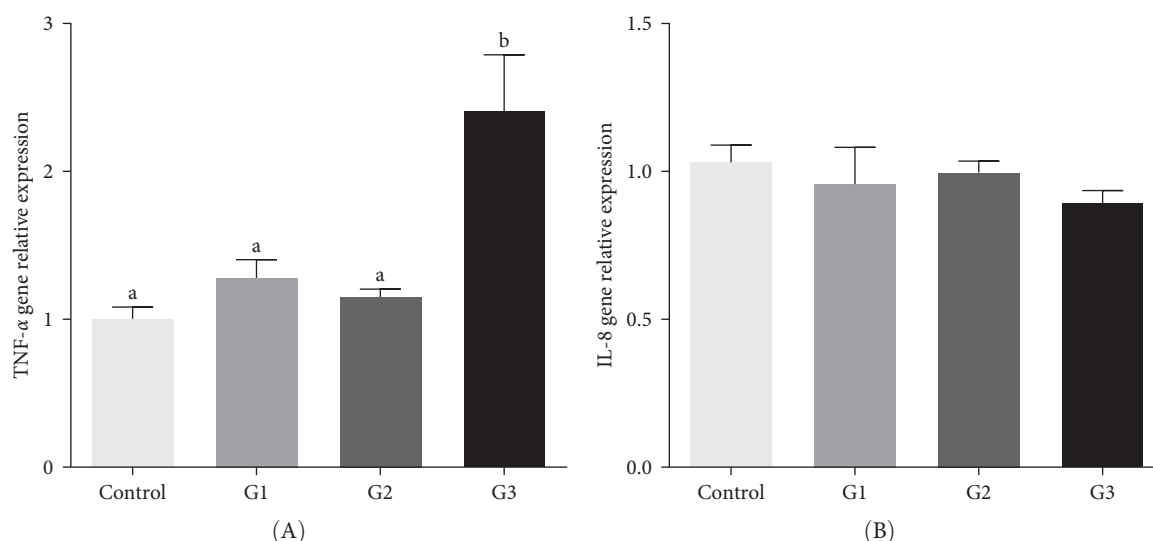


FIGURE 7: Gene relative expression of TNF- $\alpha$  (A) and IL-8 (B) in the kidney of *S. chuatsi*. Data are presented as mean  $\pm$  SD ( $n = 4$ ). Bars with different letters represent significant differences ( $p < 0.05$ , Tukey's test).

animals in the test. In this study, the addition of herbal mixtures, probiotics, or a fermented complex of both to the diets effectively improved the overall growth of *S. chuatsi* during artificial compound feed feeding. The weight gain and SGRs in G1 were significantly higher than those in the control group, indicating that adding a certain quantity of herbal mixtures to the diet could promote *S. chuatsi* growth. Our findings are corroborated by previous studies on herbal additives. The addition of a certain quantity of *H. cordata*, *E. ulmoides*, or *E. ulmoides* leaf extract and *T. mongolicum* extract to basal diets can improve the growth of grass carp and carp [21–23]. However, no detailed studies have explored the components of *H. cordata* and *T. mongolicum* to assess whether they can promote fish growth. Their unique odor produces a predatory effect, and they are rich in amino acids and other nutrients that promote fish growth, conferring the advantage of using fish herbs. *E. ulmoides* can promote fish growth by improving the palatability of the diet and richness of various amino acids, trace elements, and other nutrients [24]. Although the growth of fish in the herbal group was significantly better than that in the remaining groups, their SR was relatively low, compared with those in the Chinese herbal medicine compound probiotic group (CHCP) group. This may be because some herbs have certain toxicity, which can easily damage the liver and other important organs of the body [25]. Fermentation treatment with probiotic bacteria, such as *Lactobacillus* or *Bacillus*, can reduce or degrade the toxic substances contained in the herbal medicine itself [26, 27]. This could be due to the degradation of toxic substances in herbal medicines or modification of their structure through microbial decomposition and transformation, resulting in a change in medicinal properties or a reduction in toxic side effects [28]. In this study, the culture experiments were conducted in ponds in order to simulate actual culture production conditions; thus, it was not possible to maintain a constant temperature. Although the low water temperature in winter resulted in a consistently low overall *S. chuatsi*

food intake in each group, the experimental conditions for each group were consistent.

Serum TG and T-CHO levels can indirectly reflect lipid metabolism in the liver, as well as the health of the organism. Compounding herbs in the sturgeon diet can significantly reduce TG and T-CHO levels [29]. Moreover, the addition of probiotic preparations to the diets of rainbow trout or fattening pigs can reduce serum TG and T-CHO levels and improve lipid metabolism in animals [30, 31]. In this study, TG levels were significantly reduced in all experimental groups, with G3 being the most effective treatment. This may be due to the synergistic effects of the herbs and probiotics. Compound herbs have lipid-regulating components, such as flavonoids, which are abundant in *T. mongolicum* [32]. The lipid-lowering active components in herbs can be better released and enhanced through the synergistic effect of probiotics [33, 34]. In addition, probiotics can affect lipid metabolism by altering the intestinal environment and flora [35]. LZM and peroxidase are two important enzymes involved in nonspecific immunity in fish. LZM exerts antibacterial and bactericidal effects mainly by lysing bacterial cell walls [36]. CAT facilitates the breakdown of some peroxides produced during antioxidation and reduces oxidative stress damage in organisms. In the present study, there were signs of increased peroxidase activity in the experimental group; however, the difference was not significant. Moreover, the LZM activity in the experimental group was significantly higher than that in the control group, with G3 showing the best results. The ability of herbs and probiotics to increase LZM activity has been reported in previous studies [37–41]. Chinese herbal medicines are rich in various immunoactive substances, such as polysaccharides, alkaloids, flavonoids, glycosides, and organic acids. These substances enhance the immune level of fish through their immune activity in the organism [40]. These active ingredients can promote the proliferation of immune cells, such as macrophages, thus accelerating LZM synthesis and secretion and increasing its

activity. Probiotics can modulate innate immunity by regulating humoral immune responses and the expression of related immune genes [42, 43]. The *B. velezensis* GY65 used in this study was screened and isolated from the intestinal tract of *S. chuatsi*. Safety and functional evaluations of this probiotic showed that the strain has strong extracellular enzyme production capacity and can significantly increase serum LZM activity in fish [44]. The addition of *B. cereus*-fermented herbs to carp diets can trigger LZM activity in the serum [45]. Possibly, the cell walls of the herbs were ruptured by the extracellular enzymes produced by probiotic bacteria, improving the release of the active drug components and resulting in a greater effect [46].

GOT and GPT are mainly found in the liver tissue of animals. The serum levels of both aminotransferases are normally low, but they are released into the serum when the liver is damaged. Therefore, changes in serum GOT and GPT activity can reflect damage to liver cells. In this study, GOT activity was significantly higher in G1 than that in the G3 and control groups. In addition, G2 had significantly higher scores than the control group. There was no significant difference in GPT activity between the groups. Lu et al. [47] found that after adding compound herbs to the diets of sharpnose perch for 56 days, the GOT activity in serum showed an increasing trend and then decreased as the compound herb content increased. Huang et al. [48] similarly found that the addition of a certain amount of herbs to the diet may cause hepatocyte damage and increase serum transaminase activity. In contrast, another study found that the addition of fermented herbs with probiotic bacteria significantly reduced serum transaminase activity [45]. These results showed that the addition of fermented herbal preparations can reduce both GOT and GPT activities, and fermented herbs were superior to unfermented herbs.

Redox reactions are an extremely important metabolic pathway in living organisms that determine their aging and death while providing energy. Reactive oxygen species (ROS), harmful byproducts that damage cells, are produced during various metabolic reactions. ROS attack cells through oxidative stress, causing adverse reactions, such as lipid peroxidation and protein damage [49].

The antioxidant system plays an important role in coping with the oxidative damage caused by ROS. SOD and CAT, two critical antioxidant enzymes, mainly play a role in scavenging superoxide anion radicals produced by the body and peroxides produced during antioxidation, thus reducing oxidative cellular damage [50]. MDA is a product of lipid peroxidation, which is toxic to the cells and can cause liver damage and reflect the level of peroxidation in the liver [51]. When the liver is damaged by oxidative stress, the cell membrane is damaged by ROS or peroxides. Subsequently, GOT and GPT are released from hepatocytes into the serum. A significant decrease of both liver enzymes in the liver indicates damage to the hepatocytes. The addition of three herbal complexes to the *Schisandra* feed could increase CAT activity in the liver [52]. A study evaluating the effects of 13 different single or combined herbs on antioxidants in tilapia liver showed that both single and combined herbs reduced MDA

content [53]. The addition of *Bacillus* and photosynthetic bacteria to feeds significantly increases CAT activity in the liver, reduces MDA content, and increases antioxidant levels in the liver [54]. Feeding fermented herbal preparations of *Bacillus subtilis* to carp for 56 days resulted in a significant increase in CAT activity and a significant decrease in MDA content in the high-spiked group [55]. In this study, compound herbs, probiotics, and their combination significantly increased CAT activity and decreased the MDA content in the liver of *S. chuatsi*, with G3 exhibiting significantly lower MDA content than the other groups. There was no significant difference in SOD, GOT, and GPT levels among the groups, consistent with the results of previous studies [56]. Chinese herbs contain many flavonoids, phenols, polysaccharides, and many other natural antioxidants. These active substances can exert antioxidant effects in living organisms directly by scavenging oxygen radical groups or indirectly as raw materials for the synthesis of certain antioxidant substances [57]. Probiotics secrete antioxidant enzymes, vitamins, and other antioxidants and enhance the antioxidant capacity of the host by stimulating the antioxidant system and mediating the relevant antioxidant genes [58]. The synergistic application of herbal medicine and probiotics can improve the antioxidant effect on the body, probably due to the release of antioxidant components by breaking the cell wall of herbal medicines. However, the synergistic process may also produce new medicinal components or metabolites with improved antioxidant functions in living organisms [28].

As a relatively high-value aquaculture fish species, *S. chuatsi* can become more profitable in artificial feed cultures by improving feed digestibility to reduce feed costs. Digestive enzymes in fish intestines are mainly classified as proteases, amylases, and lipases. Their main role is to break down large food molecules into small molecules to promote digestion and absorption in the fish intestine. In the present study, the protease and lipase activities in the G1, G2, and G3 groups were significantly higher than those in the control group, and the amylase activity in G2 and G3 was significantly higher than that in the G1 and control groups. In a similar study, the addition of herbal powder and aqueous decoction to the base diet of juvenile sea bream increased its intestinal protease activity [59]. Feeding a compound diet supplemented with 1% herbal medicine had a significant effect on the digestive enzyme activity of hybrid sturgeon [60]. The *B. velezensis* GY65 used in this study could increase the intestinal protease and amylase activities of *S. chuatsi* [61]. A significant increase in protease and lipase activities was observed in the intestine of spotted catfish after 8 weeks of feeding with fermented extracts of Chinese herbs [62]. In addition, the active ingredients of herbs can promote the production and secretion of digestive enzymes by repairing the intestinal mucosa, promoting the development of intestinal villi, and increasing the number of interepithelial cup cells and lymphocytes while aligning them more closely [63]. Herbs contain many organic acids that can lower the intestinal pH after decomposition in the intestine, thereby increasing the activity of digestive enzymes [64]. *Bacillus velezensis* has a powerful extracellular enzyme production capacity, which is

directly responsible for the ability of probiotics to influence intestinal digestive enzyme activity [65]. Probiotics can also exert indirect influence by improving intestinal development and altering the intestinal flora composition. In this study, the best effect on digestive enzyme activity was observed in the combined G3 treatment. Probiotics and herbal medicines synergize and promote each other, and probiotics can promote the absorption and utilization of the active ingredients of herbal medicines in the intestine. Herbal medicine can play an active role in the proliferation of exogenous and intestinal probiotics, thus promoting the synthesis and secretion of extracellular enzymes.

Histomorphological indicators of the intestinal tract can reflect the intestinal health and function of animals, such as villi length, villi width, and basal thickness. A well-developed intestine tends to have a stronger digestive and absorptive capacity [66, 67]. In this study, all experimental groups showed better intestinal development than that in the control group. The length and basal thickness of the villi in G1 were significantly greater than those in the other groups, consistent with the growth performance results. A similar study showed that feeding Nile tilapia with different proportions of *Ginkgo biloba* extract increased the intestinal villi height, width, and muscle layer thickness [68]. One study found that adding *B. subtilis* to feed improved the morphological development of the intestinal tract of eels. [69]. Feeding *Astragalus* and *Lactobacillus* phylogeny fermenters can increase the height of carp intestinal villi and improve intestinal morphology, thereby promoting fish growth [70].

TNF- $\alpha$  is an important pro-inflammatory cytokine with a wide range of biological functions. It plays a vital role in apoptosis, tumor formation, immune regulation, and stress response and is an indispensable immunomodulatory factor for the body to maintain homeostasis and resist external stress [71]. IL-8 is mainly produced by monocytes and macrophages, and it is an early inflammatory factor with a strong chemotactic effect. It is primarily involved in inflammatory and immune responses through chemotaxis in neutrophils and T cells, and it enhances the anti-infective capacity of target cells [72]. In this study, the relative expression of the two immune-related genes in the spleen of the fish across the experimental groups was elevated, relative to that in the control group. The relative expression of TNF- $\alpha$  in the kidneys of the fish in G3 was significantly greater than that in the other groups, and there was no significant difference in IL-8 relative expression between the groups. Most current studies have shown that herbal polysaccharides, glycosides, and alkaloids are the main components of herbal medicines that affect immunity [73]. Herbal polysaccharides can exert immunomodulatory effects by activating various immune cells, such as macrophages and T cells, and by inducing the release of cytokines, such as interleukins and pro-inflammatory factors [74]. Xu and Chen [75] found that *Astragalus* polysaccharides enhanced immune and antitumor effects by promoting TNF- $\alpha$  secretion in mice. Wang [76] showed that adding compound Chinese herbs to the diet could increase the gene relative expression of TNF- $\alpha$  and IL-1 $\beta$  in five tissues, including the liver and spleen of Jiffy Tilapia. The *B. subtilis* GY65 used

here promoted the relative gene relative expression of TNF- $\alpha$  and IL-1 $\beta$  in a previous study [61]. One study found that adding *B. subtilis* to the feed significantly reduced the relative gene relative expression of the intestinal pro-inflammatory factors TNF- $\alpha$ , IL-8, IL-1 $\beta$ , and IL-15 in juvenile largemouth bass [77]. The differences between previous research and the present study may be due to differences in fish species, tissue sites, species of bacteria, as well as the effects of different experimental conditions. Notably, because of the experimental conditions and time schedule, virus attack experiments were not conducted, and the available data do not present the antiviral effect of herbs on fish. This research direction requires further exploration in the future. In addition, although the relative expression of immune-related genes in the spleens of fish was elevated in all experimental groups, there was not much difference between the G1 and G3 groups in promoting immunity in fish. This may be due to the fact that after a period of feeding, the herbs would positively regulate the inherent intestinal flora in the fish [78]. At the same time, the inherent intestinal flora would further promote the release of the immunological active ingredients of the herbs [56].

## 5. Conclusion

In conclusion, this study found that CCHM at 12.5 mg/mL had significant blocking and direct killing effects on LMBV and could be a potential “antiviral drug.” CCHM after fermentation with *B. velezensis* GY65 could effectively protect the growth and immunity of *S. chuatsi* cultured using artificial compound feed. Moreover, this study provides a theoretical basis for the practical application of herbal medicines and probiotics in aquaculture.

## Nomenclature

|                      |                                                  |
|----------------------|--------------------------------------------------|
| CAT:                 | Catalase                                         |
| CCHM:                | Compound Chinese herbal medicine                 |
| CHCP:                | Chinese herbal medicine compound probiotic group |
| CPE:                 | Cloud point extraction                           |
| EPC:                 | Endothelial progenitor cells                     |
| TCID <sub>50</sub> : | 50% tissue culture infective dose                |
| GOT:                 | Glutamic oxaloacetic transaminase                |
| GPT:                 | Glutamic pyruvic transaminase                    |
| LMBV:                | Largemouth bass virus                            |
| LZM:                 | Lysozyme                                         |
| MDA:                 | Malondialdehyde                                  |
| ROS:                 | Reactive oxygen species                          |
| SOD:                 | Superoxide dismutase                             |
| T-CHO:               | Total cholesterol                                |
| TG:                  | Triglycerides.                                   |

## Data Availability Statement

All data that support this study are available from the corresponding author upon reasonable request.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Author Contributions

Jiang-Yi Tang participated in experiments and wrote the manuscript. Xue-Zhi Zhu, Yue Wang, and Li-Yan Liu participated in experiments. Jian-Chou Chen, De-Feng Zhang, Qing Wang, and Ying-Ying Wang contributed to experimental design and guidance. Cun-Bin Shi contributed to funding acquisition, experimental design, and guidance. Ya-Jun Wang contributed to review of manuscript text, funding acquisition, experimental design, and guidance, manuscript submission, and management.

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